

Sajad Abavisani

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No visa sponsorship required

SUMMARY

Ph.D. in Electrical and Computer Engineering from Georgia Tech (completed Aug 2026), specializing in causal structure learning, large-scale graph algorithms, and LLM/agent systems. Published at ICLR 2023 (Oral) and NeurIPS 2026; scaled causal graph learning from 6-node toy problems to 100+ node graphs; built and operate production agentic pipelines on local LLMs; released open-source ML tooling (gunfolds). Looking to build trustworthy AI systems end to end, from data ingestion and modeling to deployment and evaluation.

EDUCATION

Ph.D. in Electrical & Computer Engineering

Aug 2020 – Aug 2026

Georgia Institute of Technology, Atlanta, GA — GPA 3.8/4.0

Focus: Causal Discovery, Graph Algorithms, Time Series, Deep Learning, Optimization

Coursework: Mathematical Foundations of ML, ML with Limited Supervision, Conversational AI, Convex Optimization, Statistical ML, Information Theory, Random Processes

M.S. in Electrical & Computer Engineering

Aug 2020 – Aug 2023

Georgia Institute of Technology — GPA 3.9/4.0

B.S. in Electrical Engineering

Sep 2014 – Sep 2019

Ferdowsi University of Mashhad, Iran — GPA 4.0/4.0

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, R, Java, SQL, Clingo/ASP

ML & Deep Learning: PyTorch (CUDA), TensorFlow/JAX, Hugging Face (Transformers, Accelerate, Datasets), LLM fine-tuning (LoRA/PEFT/QLoRA), RAG pipelines (LangChain, LlamaIndex), agentic AI workflows (multi-agent LLM councils, OpenClaw agent runtime), local LLM serving (Ollama: Qwen3.5-122B, Gemma 4), FastAPI, chain-of-thought reasoning, Graph Neural Networks (PyTorch Geometric), Transformers, Graph Transformers, Set Transformers, CNNs, attention mechanisms, encoder-decoder architectures, GANs, Diffusion Models, Capsule Networks, Scikit-learn

Data & Infrastructure: Pandas, NumPy, SciPy, NetworkX, Statsmodels, Docker, GCP, AWS, Azure ML, W&B, MLflow, Git, GitHub Actions CI/CD, Linux, SLURM clusters, performance profiling & optimization

Specializations: Causal inference & discovery, constraint programming (ASP/Clingo), graph algorithms (SCC decomposition, structural constraints), multi-source network integration, time series analysis & forecasting, statistical modeling, Bayesian inference, optimization, open-source development (gunfolds, llm-causal-select), agentic coding (Claude Code, Cursor)

RESEARCH EXPERIENCE

AI-Augmented Causal Discovery from Noisy Time Series | Georgia Tech

2024 – Present

- Engineered RnR, a meta-solver that refines the output graph of any causal discovery algorithm by modeling undersampling effects via ASP, improving F1 scores by 45% over SOTA baselines (PCMCI, FASK, MVGC, GIMME). Accepted at NeurIPS 2026.
- Designed prioritized multi-stage optimization (density → latent structure → edge orientation), cutting the search space by ~4 orders of magnitude while preserving global optimality guarantees; validated end to end on real fMRI data.
- Building an ML classification pipeline (healthy controls vs. schizophrenia) using Graph Transformers (BrainNetTransformer) and a novel Solution-Set Transformer that learns diagnostic biomarkers from RASL's multi-graph output — a capability unique to our method.
- Developing an LLM-guided framework that selects ground-truth causal graphs from equivalence classes by combining chain-of-thought reasoning with domain knowledge; built the LoRA/QLoRA fine-tuning and evaluation pipeline (Hugging Face PEFT; accuracy, edge-F1, SHD) and benchmarked it on simulated graphs with known ground truth.

Scalable Causal Discovery & Network Integration | Georgia Tech

2022 – 2024

- Scaled graph structure learning from 6-node toy graphs (prior SOTA) to structured graphs with 100+ nodes by exploiting strongly-connected-component (SCC) decomposition, achieving linear scaling in the number of SCCs and enabling practical neuroimaging applications for the first time.
- Ran distributed experiments on a 19-machine SLURM cluster (64 cores, 512 GB RAM each) with Clingo in 10-thread parallel mode.
- Developed ION-C, an ASP algorithm that ingests overlapping datasets measuring different variable sets and integrates them into one consistent causal graph. Proved soundness and completeness; scaled from 4–6 nodes (prior limit) to 25+ nodes. Validated on European Social Survey data. Under review at ICLR 2027.

Constraint-Based Causal Structure Learning (GRACE-C) | Georgia Tech

2020 – 2023

- Reformulated the rate-agnostic structure learning problem into a declarative ASP encoding, achieving ~10,000× speedup (17 hrs → 6 sec). **ICLR 2023 Oral (top 25%)**.
- Proved correctness and completeness via direct encoding theorem; verified against 1,000 test graphs with identical outputs to provably correct RASL baseline.
- Built optimization mode with weighted constraint relaxation for noisy inputs; applied to real resting-state fMRI.

- Co-developed dRASL for deliberate undersampling: proved that measuring *more slowly* can reduce causal uncertainty by up to 4 orders of magnitude. Published at CLeaR/PMLR 2023.
- Released all methods as the pip-installable gunfolds open-source package with CI/CD and Sphinx documentation. (pypi.org/project/gunfolds)

Deep Learning for Computer Vision & Medical Imaging | Georgia Tech / Rutgers

2019 – 2021

- Co-developed DECAPS for COVID-19 CT classification: 96.1% AUC with inverted dynamic routing and conditional GAN augmentation (pix2pix), outperforming three board-certified radiologists (95% vs 85% TPR at matched FPR).
- Co-authored Greedy AutoAugment: reduced augmentation policy search from exponential to linear complexity, 360× fewer GPU-hours than AutoAugment. Published in Pattern Recognition Letters.

Robotic Prosthetic Control | Ferdowsi University

2017 – 2019

- Designed multimodal neural networks (EMG + vision) for prosthetic gesture recognition using CNNs and restricted Boltzmann machines.

SELECTED INDUSTRY-RELEVANT PROJECTS

- **Local LLM Council with API Chairman (Jul 2026):** Extended karpathy/llm-council into a hybrid council: two local models on Ollama (Qwen3.5-122B, Gemma 4) answer independently, then anonymously review and rank each other's responses, and a closed-source chairman (GPT-5.5) synthesizes the final decision. Served as an OpenAI-compatible FastAPI endpoint so any agent framework can call the full three-stage deliberation as a single model; registered as a model provider in OpenClaw.
- **Daily Automated Market-Research Pipeline (Jul 2026 – present, runs every trading day):** Built and operate an agentic pipeline that researches a live equity portfolio before each market open: keyless ingestion of quotes and news (yfinance, Google News RSS); council filtering in which Gemma 4 triages every article and Qwen3.5-122B re-judges survivors, keeping only unanimous “material” verdicts; per-holding briefs; earnings-date extraction to Google Calendar; delivery by email and Telegram. Seven checkpointed, fixed-context stages keep every LLM call inside a local context window and make runs resumable after crashes; NYSE-holiday aware; scheduled with OpenClaw cron on a headless Apple-silicon server; holdings never leave the machine.
- **Local-LLM Portfolio Research Agent (Jun 2026):** Ran a fully local agent (Qwen3.5-122B on Ollama through OpenClaw, with web-search tools) to research a 40-position equity portfolio: per-holding reports (business model, valuation multiples, catalysts, risks, analyst targets), a portfolio-level sector and concentration review, and a 40-to-15 position rebalancing roadmap under a value-based growth framework, published as a linked Obsidian wiki.
- **LLM-Guided Causal Graph Selection:** Fine-tuned LLMs (LoRA/QLoRA; Mistral-7B, Llama-3-8B) with chain-of-thought prompting to select ground-truth DAGs from equivalence classes, integrating domain knowledge with data-driven reasoning.
- **Solution-Set Transformer for Brain Classification:** Designing a two-level architecture (graph encoder + Set Transformer + classifier) that learns diagnostic biomarkers from multiple candidate causal graphs per subject — uniquely enabled by RASL's multi-solution output.
- **Net Worth & Portfolio Estimator:** Built a Python tool that ingests Yahoo Finance API data (yfinance), brokerage CSV exports, and real-estate valuations into a unified portfolio and net-worth model with per-position breakdown.
- **iOS Accent-Learning App:** Swift app leveraging OpenAI Whisper for speech recognition with quantitative pronunciation feedback.
- **LLM + Knowledge Graph Reasoning:** Integrated Llama 3 with knowledge graphs to improve logical reasoning on riddle-solving tasks.
- **GAN Robustness for Autonomous Driving:** Trained GANs to augment street-sign datasets, improving classification under adverse weather using multimodal inputs.

PUBLICATIONS

M. Abavisani, D. Danks, S. Plis. *GRACE-C: Generalized Rate Agnostic Causal Estimation via Constraints*. ICLR23 (**Oral, Top 25%**)

M. Abavisani et al. *RnR: A Meta-Solver for Causal Discovery in Undersampled Time Series*. **NeurIPS 2026**.

K. Solovyeva, D. Danks, **M. Abavisani**, S. Plis. *Causal Learning through Deliberate Undersampling*. CLeaR / PMLR 2023.

P. Nair et al. (incl. **M. Abavisani**). *ION-C: Integration of Overlapping Networks via Constraints*. Under review, ICLR 2027.

A. Naghizadeh, **M. Abavisani**, D. Metaxas. *Greedy AutoAugment*. Pattern Recognition Letters 2020.

A. Mobiny et al. (incl. **M. Abavisani**). *Radiologist-Level COVID-19 Detection via Capsule Networks*. arXiv 2020.

HONORS

- ICLR 2023 Oral Presentation (Top 25% of accepted papers)
- Ranked 3rd in ACM Programming Competitions
- Top 0.001% (537th / 1,000,000+) in nationwide university entrance exams, Iran
- Accepted into NODET (National Organization for Development of Exceptional Talents)